

An Anonymous Authentication Scheme for Internet of Vehicles Based on TRUG-PBFT Main–Secondary Chains and Zero-Knowledge Proof

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Abstract—The application of Internet of Vehicles (IoV) technology has greatly improved users' driving experience, but it also faces some challenges: 1) the central server is not powerful enough to support the rapid growth of IoV identity authentication requests and 2) there is a privacy leakage issue during vehicle authentication. To address these issues, we propose an anonymous authentication scheme based on trustworthy roadside unit group (TRUG)-PBFT main secondary chains and zero-knowledge proof (ZKP). First, to enhance authentication efficiency, we propose the TRUG-PBFT consensus algorithm. It improves the traditional PBFT by optimizing the PBFT consensus process, reducing the number of consensus nodes using fractional grouping, and selecting main node using verifiable random functions (VRFs). Second, we use a lattice-based ZKP scheme to achieve anonymous authentication of vehicles, and important data in the vehicle authentication process is stored by the main chain maintained by the base station group and the secondary chain maintained by the roadside unit group. Finally, experimental results demonstrate that compared to PBFT consensus, TRUG-PBFT in terms of consensus efficiency is improved by approximately 33%, and the authentication scheme's computational cost is only 7.08 ms, superior to existing authentication schemes.

Index Terms—Anonymous authentication, blockchain, consensus algorithm, Internet of Vehicles (IoV), zero-knowledge proof (ZKP).

I. INTRODUCTION

THE Internet of Vehicles (IoV) has emerged as a novel technology supporting autonomous and intelligent driving, and enhancing transportation services and operational efficiency [1]. It signifies the automotive industry's ongoing transition toward intelligence and interconnectivity. Through the integration of advanced sensor devices and intelligent control systems, vehicles leverage wireless network communication technologies to enable five distinct communication scenarios: vehicle-to-cloud, vehicle-to-vehicle, vehicle-to-road, vehicle-to-human, and intravehicle communication.

The IoV brings remarkable convenience to people's lives, but it also presents significant threats to users' security

and privacy [2]. Due to the diversity of applications and devices within the IoV, it has become a primary target for network attacks. These widespread attacks can lead to severe security concerns. For instance, sensitive information like customer identities, vehicle speeds, and locations may be collected within IoV services [3]. Once exposed, sensitive data of legitimate drivers could potentially lead to accidents through the manipulation of unsafe route planning services by malicious attackers. Hence, anonymity is paramount. If an attacker can identify user identities by linking them through authentication contexts, it poses a severe threat to user privacy and vehicle security [3]. Authentication is one of the fundamental security mechanisms in IoV systems to prevent identity forgery and illegal access, yet most of them lack anonymous features. To address these issues, some solutions attempt to authenticate users using anonymous methods. However, the direct application of these methods is impractical due to the additional overhead caused by complex cryptographic calculations. Traditional authentication methods based on anonymous credentials [4] provide robust anonymity, unlinkability, and traceability. However, these methods involve complex algorithms, such as bilinear mapping and group signatures, potentially imposing significant computational burdens on vehicles. Pseudonym-based approaches [5] cannot guarantee the unlinkability of vehicle users, and prolonged use of pseudonyms might expose their real identities. Additionally, these methods require frequent pseudonym updates for enhanced security, adding extra burdens to vehicles. Zero-knowledge proof (ZKP) allows a prover to showcase their identity without divulging any useful information to the verifier. Current anonymity methods based on ZKP [6], [7] are efficient without requiring complex security algorithms. Users can mutually authenticate each other without the involvement of a third party.

Furthermore, traditional IoV authentication systems are typically linked by a central server, a structure often found in open wireless network environments with rapidly changing topologies. While this structure is straightforward and facilitates the control of terminal devices, a vulnerability arises if the central server comes under attack, potentially causing the entire system to become paralyzed [8]. Additionally, as the network scales, the burden on the central server increases. This might lead to amplified network latency and even network collapse. Blockchain technology has shown excellent performance in identity authentication security, presenting itself as an optimal

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solution for IoV systems [9]. The distributed structure of blockchain reduces server stress and the risk of paralysis [10], making it suitable for addressing issues within IoV systems. The consensus algorithm stands as a core method to ensure the consistency of blockchain data. The Paxos algorithm is highly favored for its fault-tolerant mechanism [11], needing only $2f+1$ nonfaulty nodes to guarantee system fault tolerance while maintaining data consistency, avoiding issues resulting from node failures or message loss. Nevertheless, the Paxos algorithm suffers from a potential deadlock scenario caused by rejected proposals, leading to a loop. Another widely used consensus algorithm is the PBFT algorithm, which achieves consensus through node voting, typically concluding within 2–5 s. For IoV, shorter consensus time reduces the difficulty of ensuring vehicle safety; however, the PBFT algorithm has scalability issues [12].

Therefore, in order to address user privacy protection and single point of failure issues, while also considering communication overhead and latency, this article proposes an IoV anonymous authentication scheme based on trustworthy roadside unit group (TRUG)-PBFT main-secondary chain and ZKP. By improving the PBFT consensus algorithm, communication overhead is reduced, authentication efficiency is improved, and ZKP is employed to safeguard user privacy during the authentication process. The contributions of this article are as follows.

First, in response to the issues of low efficiency and poor scalability of consensus algorithms, we propose the TRUG-PBFT consensus algorithm. Nodes are divided into consensus group and candidate group based on scores, and a verifiable random function (VRF) is used to select the primary node in the consensus group. Compared to the original PBFT algorithm, this algorithm has lower communication overhead and higher throughput. Therefore, it is more suitable for IoV systems.

Second, to address user privacy concerns, we propose a lattice-based ZKP distributed authentication algorithm. Proof values are generated by vehicles, computed by trusted roadside units (RSUs), and verified by base stations (BSs), with the authentication process data stored on the blockchain. Additionally, to enhance the scalability of the verification strategy, we establish a main chain maintained by a group of BSs and a secondary chain maintained by RSUs. Through interaction between the main chain and the secondary chain, a trust bridge is built between vehicles and BSs, ensuring the privacy and security of vehicles.

Finally, through experimentation, we compare our consensus algorithm with existing consensus solutions in terms of throughput, latency, and communication overhead. Additionally, we calculate the computational cost of the authentication scheme and compare it with existing ZKP-based authentication schemes in terms of authentication cost. Subsequently, to validate the practicality of the solution in the real world, we simultaneously consider both the consensus overhead and authentication cost, conducting simulation experiments. The results indicate that our approach is rational, practical, and suitable for deployment in the IoV environment.

II. RELATED WORK

To safeguard user privacy from unlawful collection, many researchers have proposed various anonymous authentication schemes (as shown in Table I). They employ different encryption methods to support user anonymity, mainly including pseudonyms, certificateless signatures, and ZKP.

Establishing pseudonym-based system enables vehicles to communicate effectively with others anonymously. However, to preserve privacy, vehicles need to frequently change these pseudonyms. In this process, vehicles either need to communicate with certificate authorities, incurring high computational and communication costs, or they must preload a large number of keys/certificates, necessitating high-capacity data storage. Moreover, pseudonym-based approaches require maintaining a tracking list to prevent tracking of malicious or compromised vehicles, resulting in significant resource wastage. In response to these issues, existing research has proposed some solutions. Nath and Choudhury [5] proposed a mutual authentication method for group communication in vehicular ad-hoc networks (VANETs), which does not require pseudonym regeneration during authentication and message exchange. Meanwhile, utilizing group keys, vehicles can communicate and authenticate with other vehicles within the group. However, this scheme fails to meet the security requirements of traceability and is vulnerable to man-in-the-middle (MITM) attacks. Qi et al. [13], while ensuring privacy in VANET conditions, combined pseudonymous mechanisms and certificateless signature mechanisms to construct an anonymous authentication scheme. Additionally, this scheme has significant advantages in terms of rapid revocation of vehicles and ensures that pseudonyms cannot be linked before tracing. Security analysis demonstrates that the scheme can meet the privacy protection requirements in VANETs. However, the scheme involves complex bilinear pairings and is not suitable for the IoV environment.

Certificateless signatures are a viable encryption tool, offering data integrity and identity authentication. They eliminate the cumbersome certificate management found in certificate-based signature systems and address the key escrow problem inherent in identity-based cryptosystems. Samra and Fouzi [4] proposed a novel framework, the certificateless aggregate signature based on traceable ring signature (CLA-TRS). This scheme aims to ensure privacy-preserving identity verification in VANET communication. Compared to existing solutions, it boasts higher computational efficiency, employing constant pairing operations that are more effective in the verification process than other schemes requiring more pairing operations. Zhou et al. [14] introduced an efficient certificateless conditional privacy-preserving authentication (CLCPPA) scheme tailored for VANETs, devoid of complex bilinear pairings and MapToPoint operations. Performance evaluation against other existing schemes revealed its efficiency, with improved efficiency in signature generation (up by 66.75%), verification (up by 33.19%), and reduced bandwidth requirements (down by 14.75%) compared to previous state-of-the-art solutions. However, most existing certificateless signature schemes are not immune to various attacks, such as public-key replacement

TABLE I
EXISTING RESEARCH PLANS

Research	Concept applied	Advantages
Nath <i>et al.</i> [5]	*Elliptic curve cryptography (ECC); *Group key	*No need for repetitive pseudonym generation; *RSUs authenticate vehicles anonymously, while vehicles authenticate and communicate with each other using group keys.
Qi <i>et al.</i> [13]	*Elliptic curve discrete logarithm problem (ECDLP); *Bilinear pairing; *(k, n)-CAA problem; *Fiat-Shamir heuristic	*Avoided certificate management and key escrow; *Provided self-generation mechanism for secure and efficient pseudonym replacement
Samra <i>et al.</i> [4]	*Diffie-Hellman problem; *Traceable ring signature	*Constant pairing operations, reducing computational overhead and communication latency for signature verification; *Proof of existential unforgeability, anonymity, and traceability
Zhou <i>et al.</i> [14]	*ECDLP;	*No need for any complex bilinear pairing and MapToPoint operations;
Rasheed <i>et al.</i> [6]	*Zero-knowledge proof based on quadratic residues; *Based on symmetric cryptographic systems	*Users can select privacy levels based on the service; *RSUs can revoke permissions from authorized vehicles
Xi <i>et al.</i> [7]	*Fujisaki-Okamoto commitment; *ECC	*During the verification process, anonymity, unlinkability, and traceability are ensured; *A precomputation mechanism is employed, eliminating the burden of complex computations; *A fast reconnection protocol is designed
Yao <i>et al.</i> [15]	*Diffie-Hellman problem; *ECDLP; *Blockchain	*Flexible cross-datacenter identity authentication; *Implementing anonymity to protect vehicle privacy
Shen <i>et al.</i> [16]	*Elliptic curve digital signature algorithm (ECDSA); *Blockchain	*Mutual authentication between vehicles, edge nodes, and cloud servers has been achieved; *Re-authentication process incurring minimal overhead
Shahrouz <i>et al.</i> [17]	*Zero-Knowledge succinct non-interactive argument of knowledge (zk-SNARK); *Blockchain	*The repetition of computations in storing proofs on the blockchain reduces authentication costs
Xu <i>et al.</i> [18]	*ECC; *PBFT consensus	*Optimization of the PBFT consensus process; *Reducing the number of consensus nodes through fractional grouping

attacks or malicious-but-passive key-generation-center attacks. Moreover, computational complexity and high communication costs are also significant considerations.

ZKP offers a promising solution to the aforementioned challenges. While ensuring user anonymity, they incur smaller computational, communication, and storage costs compared to other authentication methods, meeting the security requirements of the IoV environment. Rasheed *et al.* [6] implemented a ZKP-based authentication system for IoV. The server can authenticate users without revealing any identities, and the overhead of ZKP algorithms is minimal. However, this approach requires preshared keys. Xi *et al.* [7] devised an IoV anonymous mutual authentication protocol based on ZKP and elliptic curve cryptography (ECC). It utilizes lightweight Fujisaki-Okamoto (FO) commitments and ECC for efficient mutual authentication. Additionally, to address reconnection issues, a rapid reconnection protocol is designed based on the vehicles' recent access security context (i.e., session keys and proofs). Nevertheless, most current ZKP-based IoV identity authentication relies on central servers. As network scales, the burden on central servers increases with the growing volume of authentication information, leading to heightened network latency or even network crashes.

Due to the distributed nature of blockchain, which mitigates server bottlenecks and crashes, coupled with its tamper-resistant and immutable properties, leveraging blockchain advantages to address issues in the IoV system is highly suitable. Yao *et al.* [15] proposed a lightweight anonymous

identity verification mechanism based on blockchain for distributed IoV services. They achieved cross-data center authentication by combining modern cryptographic techniques with blockchain technology. Shen *et al.* [16] introduced a secure and efficient blockchain-based IoV identity verification protocol named SEA. This protocol facilitates mutual authentication among vehicles, edge nodes, and cloud servers. Furthermore, the cloud server only engages in initial identity verification when unverified vehicles access its services for the first time. Subsequent verifications are performed by edge nodes by checking the authentication results recorded on the blockchain, thus minimizing cryptographic computations and communication overhead. Shahrouz and Analoui [17] introduced simulated extractable zero-knowledge succinct noninteractive argument of knowledge (SE zk-SNARKs) into the IoV environment. They separated the critical components of proof statements and stored them in the blockchain as immutable data storage to adapt to the high real-time nature of IoV and eliminate redundant computations, making the proposed scheme lightweight enough to operate efficiently on VANETs. Xu *et al.* [18] proposed a blockchain-based identity authentication scheme and improved the PBFT consensus algorithm, introducing a consensus algorithm called SG-PBFT. This algorithm is more efficient than the original PBFT algorithm, with less communication overhead and higher throughput, making it more suitable for the IoV environment.

Currently, blockchain-based authentication schemes are affected in terms of verification time by two main factors:

1) the authentication algorithm itself and 2) the efficiency of the blockchain consensus algorithm. However, previous solutions have only focused on one of these aspects. Therefore, this article aims to investigate both aspects to improve the PBFT consensus algorithm, enhance its efficiency, and integrate it with ZKP algorithms to collectively enhance authentication efficiency.

III. SYSTEM STRUCTURE

A. Network Model

IoV data contains personal privacy information that cannot be accessed without authorization. Therefore, both the main chain and secondary chains in this article adopt a consortium chain structure. All users and organizations joining the blockchain network must be authorized by the traffic management department. The main chain serves as a global trusted sharing platform, primarily used for cross-chain trusted authentication and interaction. Main chain nodes are constructed and maintained by public and trusted organizations, such as the government, automobile manufacturers, and service providers, and are considered trusted devices. Due to the trustworthy nature of the main chain nodes, their limited number, and slow growth rate, the main chain employs the classical PBFT consensus algorithm.

The secondary chain acts as a regional platform, primarily used for storing authentication data within the region. Secondary chain nodes are constructed and maintained by private enterprises and telecommunications operators. Although these organizations are trustworthy, secondary chain nodes (such as RSUs) are visibly distributed along roads, making them susceptible to attacks or disruptions, and are therefore considered semi-trusted devices. Given the semi-trusted nature, large number, and rapid growth of secondary chain nodes, the secondary chain employs the TRUG-PBFT consensus algorithm proposed in this article. Since the main chain needs to store global information, the secondary chain's primary nodes package the hash values and remarks of the recorded blocks in their respective regions into transactions broadcasted to the main chain nodes. In this article, the main chain nodes are mostly BSs equipped with high-performance servers, possessing high computing power and vast data storage space. Secondary chain nodes are mostly RSUs, equipped with edge servers with moderate computing power and storage capacity.

The dynamic security authentication architecture for the IoV in this article comprises a main chain and several secondary chains [19], as illustrated in Fig. 1. We primarily focus on four entities: 1) blockchain network; 2) BS; 3) RSUs; and 4) vehicles.

Blockchain Network: Composed of a main chain and several secondary chains, the blockchain network serves as a peer-to-peer network with crucial attributes of immutable, nonrepudiable, and verifiable data storage. Secondary chains record the time when vehicles are initially validated. When vehicles interact with RSUs within the service range of BSs for the second time, secondary chains determine whether revalidation is necessary based on the time of the last successful validation. When the time interval exceeds a predefined

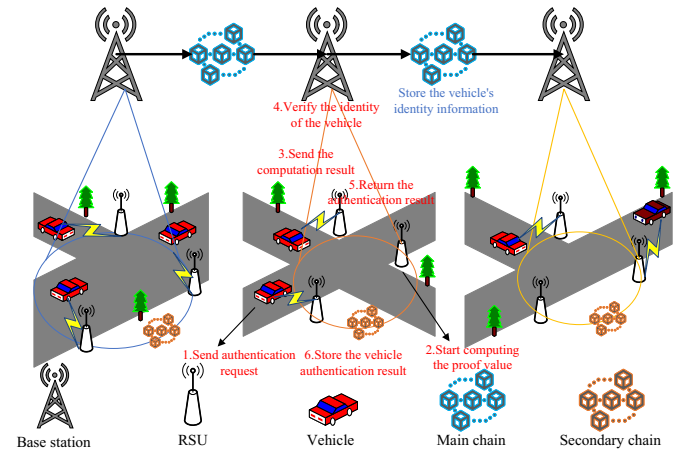


Fig. 1. Network model.

threshold, identity verification and authorization are initiated. The main chain stores the privacy information of vehicles and provides it to the secondary chains in the form of hash values when vehicle identities need to be verified. Additionally, when vehicles move across BSs, the stations can query verification records from the main chain to delay the verification process, ensuring continuity in vehicle tasks.

BS: In the IoV environment, areas are divided based on geographical locations, with each area having a BS. Serving as trusted entities, BSs maintain the main blockchain and store the privacy information of vehicles. They verify vehicle identities during the authentication process and then return the verification results to RSUs.

RSUs: RSUs serve as infrastructure communication nodes, facilitating communication between vehicles and BSs. In this article, all RSUs within the coverage area of a BS are grouped into a set, and the maintenance of the secondary chain is managed by the RSUs in this set. RSUs are intuitively distributed along roads and are susceptible to attacks or disruptions by adversaries. In this article, RSUs are considered semi-trusted devices. RSUs are responsible for computing the evidence sent by vehicles during the ZKP process and for reaching consensus after receiving authentication results from BSs, disseminating the results throughout the entire set. Only vehicles that pass identity authentication are allowed to access the network, while malicious vehicles are denied access.

Vehicles: Equipped with onboard units (OBUs) capable of supporting the DSCR protocol, vehicles are generally assumed to have limited computing and storage capabilities. They do not maintain a blockchain locally or participate in the consensus process. Instead, they utilize ZKP for identity authentication, generating zero-knowledge evidence from their own privacy information and transmitting it to RSUs, then waiting for authentication results. Upon successful identity authentication, they retrieve road information or in-vehicle entertainment from the network.

B. TRUG-PBFT Consensus

1) TRUG-PBFT Grouping Mechanism: In the IoV, which encompasses a large number of moving vehicles, ensuring

efficient vehicle authentication is paramount. To guarantee the efficiency of authentication, this article divides the set of RSUs into two categories based on a comprehensive score: 1) the consensus set and 2) the candidate set. The consensus set, termed TRUG, is established to provide convenient authentication services for users. In the IoV, vehicles need to authenticate with RSUs frequently to access real-time road services. However, the high authentication overhead not only impacts the driving experience of vehicle owners but also increases computational burden. Therefore, establishing a trusted consensus set can streamline the authentication process for users. Vehicles only need to undergo reliable identity authentication with consensus nodes within the TRUG, allowing them to authenticate and pass through other nodes within the group. Although most RSUs are constructed by governments and have high credibility, the possibility of some RSUs being controlled by malicious users cannot be ruled out. Hence, this article proposes an improved PBFT algorithm, TRUG-PBFT consensus algorithm, which is efficient and supports Byzantine fault tolerance. The entire set of RSUs are divided into consensus group and candidate group, with RSUs of low credibility excluded from the consensus group to ensure the security and reliability of nodes within the group. The node information of consensus group members is stored in the form of blocks in the chain. When a vehicle requests authentication, consensus is initiated by members of the consensus group, and a consensus result is obtained. After the consensus is reached, the local states of all nodes are updated. Malicious nodes with low comprehensive scores are moved to the candidate group, while candidate nodes with high comprehensive scores are moved to the consensus group.

The security of the TRUG architecture primarily emphasizes the security of RSU nodes and vehicle privacy. For RSU security, ensuring that malicious RSUs are excluded from the consensus group assures the safety of consensus group members, maintaining access security at the device level without necessitating that all nodes be entirely legitimate and trustworthy, thereby narrowing the scope and difficulty of RSU security assurance. Regarding vehicle privacy, adopting ZKP ensures the confidentiality of vehicle authentication process data. In terms of performance, the group authentication mechanism significantly reduces the frequency of vehicle authentication and computational overheads.

2) *Dynamically Selecting Consensus Nodes*: Due to the high consensus latency and communication overhead of the PBFT algorithm in large-scale network conditions, a grouping approach is employed to further reduce the number of consensus nodes. This article divides RSU nodes into two groups: 1) the consensus group and 2) the candidate group. Moreover, to prevent malicious nodes from being elected as primary node, the node's score is used as the basis for grouping [20]. This ensures that the consensus nodes are consistently the most reliable throughout the consensus process, enabling dynamic optimization of the consensus nodes.

The formula for node credit scoring is as follows:

$$C_i = C_i + 2^0 S_i - 2^3 F_i, \quad i = 1, 2, 3, \dots \quad (1)$$

Among them, C is the credit score of the node, S represents the successful consensus of the node, and F represents unsuccessful. If the consensus is successful, the S value is 1 and the F value is 0; if it is unsuccessful, the S value is 0 and the F value is 1.

The calculation formula of node comprehensive score is

$$G_i = \frac{(C_i + G_{i-1})}{2}. \quad (2)$$

G_i represents the comprehensive score of node i , and also represents the credibility of node i . The initial credit scores and comprehensive scores of all nodes are recorded as 100 points.

3) *Selection of Primary Node*: In the traditional PBFT consensus algorithm, the primary node selection is achieved through a round-robin method, typically defined by the strategy $p = v \bmod n$, where p represents the primary node's identification number, v denotes the view number, and n stands for the total count of consensus nodes. However, in the realm of the IoV, a significant presence of malicious nodes and potential threats poses challenges to the security of the round-robin method. This is because attackers can easily predict the selection of the next primary node.

To address this issue, this study implements the use of VRF for primary node selection [21]. This new method, without altering its original purpose, provides higher security and predictability. VRF, a cryptographic primitive, generates randomness using a private key and allows all participants to verify the validity of the generated random number. Therefore, by introducing VRF, the selection of the primary node becomes more reliable, minimizing opportunities for manipulation by attackers.

When VRF is employed for primary node selection, it requires establishing a common parameter as a seed. This study opts to use the blockchain's height $blockchain_h$ as this common parameter. To ensure the safety and reliability of the primary node, it is selected from the consensus group. During the execution of the VRF process, the VRF verification information is transmitted to all nodes within the consensus group. The specific process for selecting the primary node proceeds as follows.

- 1) Each node RSU_i ($RSU_i \in \text{consensus group}$) utilizes $blockchain_h$ and the node's private key RSU_i^{Sr} as inputs for VRF to generate $hash_{\text{value}}$ and $proof_{\text{value}}$. The calculation methods for these hash and proof values are as follows:

$$hash_{\text{value}} = \text{VRF_hash}(RSU_i^{Sr}, blockchain_h) \quad (3)$$

$$proof_{\text{value}} = \text{VRF_proof}(RSU_i^{Sr}, blockchain_h). \quad (4)$$

- 2) The RSU_i node determines its eligibility as a primary node based on the generated hash value. This study employs the approach outlined in [22] to make this determination

$$\frac{hash(hash_{\text{value}})}{2^{\text{hashlen}}} > \lambda \quad (\lambda \in [0, 1]). \quad (5)$$

Here, $hash(\cdot)$ represents the hashing algorithm, $hashlen$ stands for the output length of $hash(\cdot)$, and λ is

TABLE II
SYMBOL TABLE

Symbols	Description
o	The operation requested by the client
t	Timestamp
c	Client identity proof
m	Client request content
i	Node number
v	View number
n	Message number
d	Requested hash value
s_c	Client's signature of the message
s_p	The signature of the message by the primary node
s_i	The signature of the message by node i
G_i	Comprehensive score of node i
$blockchain_h$	Blockchain height
CON	Consensus node set
CAN	Candidate node set
p	Primary node
N	Total number of nodes
$result$	Consensus result set

the threshold. If the $hash_{value}$ satisfies the aforementioned equation, the node becomes eligible to be selected as the primary node, initiating subsequent steps. If not, the $blockchain_h$ is decremented by 1, and the process is repeated until a primary node is chosen.

- 3) The RSU_i node then transmits the VRF verification message $\{hash_{value}, proof_{value}, blockchain_h\}$ to all nodes within the consensus group.
- 4) Upon reception of the VRF verification message, nodes within the consensus group conduct the following validation steps:

$$hash_{value} = VRF_P2H(proof_{value}) \quad (6)$$

$$\text{True/False} = VRF_verify(RSU_i^{Pr}, blockchain_h, proof_{value}). \quad (7)$$

RSU_i^{Pr} represents the public key of node RSU_i . If the verification outcome is true, it confirms that the hash value was generated by node RSU_i through rehashing. Simultaneously, they need to verify $(hash(hash_{value})/2^{hashlen}) > \lambda$, if it is true, it signifies that node RSU_i can be chosen as the primary node.

4) *TRUG-PBFT Consensus Process*: The main symbols used in this process and their explanations are shown in Table II. The pseudocode for the TRUG-PBFT consensus algorithm is depicted in Algorithm 1, while the optimized primary steps of TRUG-PBFT are shown in Fig. 2. It comprises seven main stages.

Step-1 (Initialization of All RSU Nodes): Each node starts with an initial score of 100 points and is randomly assigned to either the consensus node set or the candidate node set. During this initialization process, the order of nodes is randomized. In the consensus process, the consensus node set plays a critical role by executing the consensus task. Candidate nodes, however, do not directly participate in the consensus process but update their local state after consensus is reached, providing support and backup. The system comprises a total of N nodes, with a maximum allowable number of Byzantine nodes denoted as f . Among them, $2N/3$ nodes are specified

Algorithm 1 TRUG-PBFT

Input: $CON = \{con_1, con_2, \dots, con_i\}$, $CAN = \{can_1, can_2, \dots, can_j\}$, $G = \{g_1, g_2, \dots, g_k\}$, $blockchain_h, p$

Output: $CON' = \{con'_1, con'_2, \dots, con'_i\}$, $CAN' = \{can'_1, can'_2, \dots, can'_j\}$, $G' = \{g'_1, g'_2, \dots, g'_k\}$, p'

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1: while  $\langle REQUEST, o, t, c, s_c \rangle = \text{TRUE}$  do
2:   broadcast  $\langle \langle PRE - PREPARE, v, n, d \rangle_{s_p}, m \rangle$ ;
3:   if  $\langle PRE - PREPARE \rangle = \text{TRUE}$  then
4:     broadcast  $\langle \langle PREPARE, v, n, d, i \rangle_{s_i} \rangle$ ;
5:     receive  $\langle \langle PREPARE, v, n, d, k \rangle_{s_k} \rangle$ ;
6:   else
7:     do nothing;
8:   end if
9:   if  $prepared(d, v, n, i) = \text{TRUE}$  then
10:    Put the result of node  $i$  into the result set  $result\{\}$ ;
11:   else
12:    do nothing;
13:   end if
14:   if  $\exists \sum (result\{i\} \in result\{\}) > (2f + 1)$  then
15:    The primary node transmits the consensus result  $result\{i\}$  to all nodes;
16:    Each node compares its own result with  $result\{i\}$ ;
17:    Calculate the node's credit score based on Eq. (1);
18:    Computing the node's comprehensive score according to Eq. (2);
19:    Update the  $G$  set;
20:   else
21:    do nothing;
22:   end if
23:    $request\_count++$ ;
24:   if  $request\_count = 100$  then
25:    Add the top  $N/3$  nodes based on their comprehensive scores to set  $CON'$ ;
26:    Add the remaining nodes to set  $CAN'$ ;
27:    From set  $CON'$ , select the primary node  $p'$  using the algorithm outlined in the previous section;
28:     $request\_count = 0$ ;
29:   end if
30: end while
31: return  $(CON', CAN', G', p')$ ;
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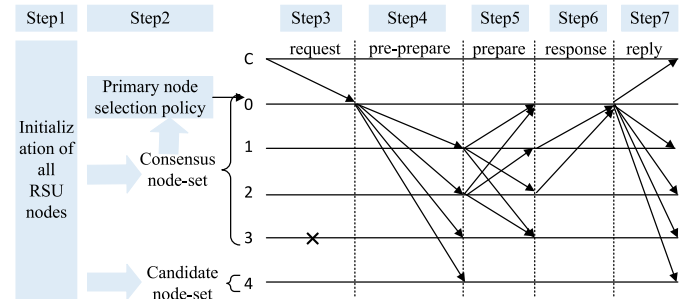


Fig. 2. TRUG-PBFT algorithm process.

as candidate nodes, and the number of consensus nodes needs to fulfill the condition $N/3 \geq 3f + 1$ to ensure the system achieves consensus under any circumstances.

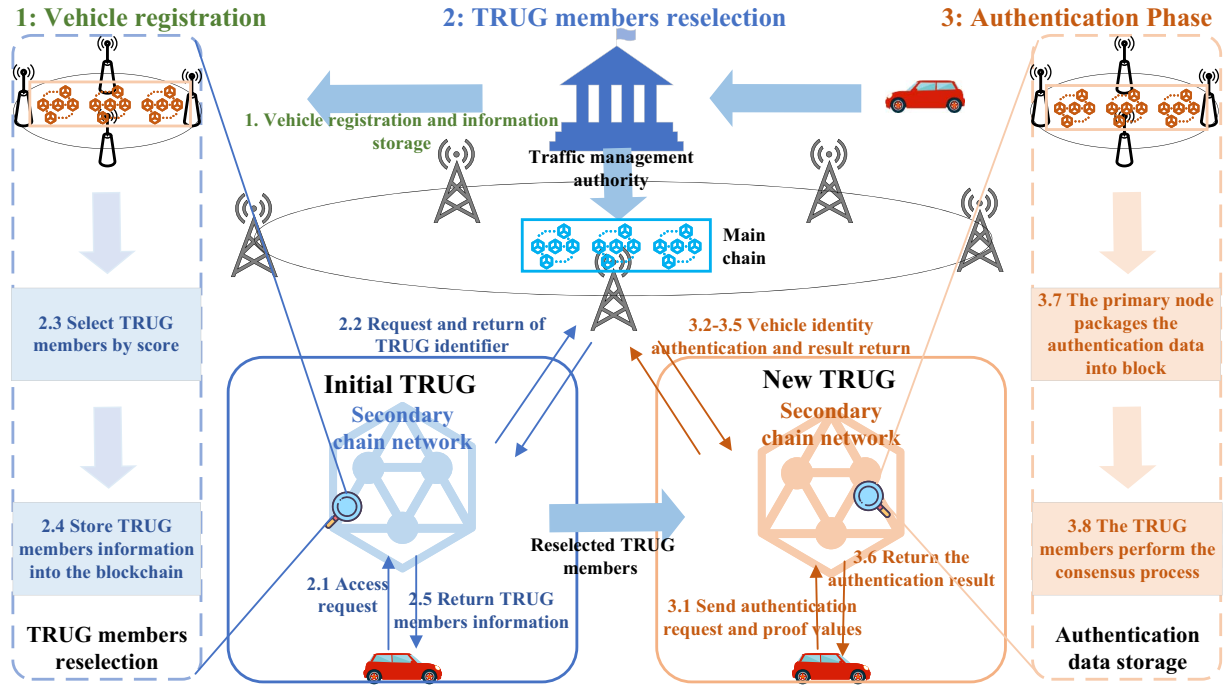


Fig. 3. Overall architecture of the solution.

Step-2 (Primary Node Selection Policy): The primary node selection policy is defined as VRF. In the event of an attack or failure of the primary node, the view will transition, and the primary node will rotate every 20 requests.

Step-3 (Client Request Phase): The client node initiates a request message to the primary node in the form of $\langle \text{REQUEST}, o, t, c, s_c \rangle$.

Step-4 (Pre-Prepare Phase): Upon receiving the request from the client, the primary node enters the pre-prepare phase. It broadcasts the message $\langle \langle \text{PRE} - \text{PREPARE}, v, n, d \rangle_{s_p}, m \rangle$ to all nodes (including candidate nodes). Sending message to candidate nodes aims to assess if the node's confirmation aligns with the consensus result, facilitating the update of candidate node scores. However, candidate nodes do not proceed to subsequent stages.

Step-5 (Prepare Phase): When a replica node receives the pre-prepare message, it verifies the message. If the verification is successful, it enters the prepare phase. If verification fails, no further action is taken. Upon entering the prepare phase, the replica node broadcasts the prepare message $\langle \langle \text{PREPARE}, v, n, d, i \rangle_{s_i} \rangle$. As a replica node sends a prepare message, it also receives and verifies prepare messages broadcasted by other nodes.

Step-6 (Response Phase): If a node receives over $2f + 1$ prepare messages, it proceeds to the commit phase, verifying the messages and providing the results to the primary node.

Step-7 (Reply Phase): Once the primary node receives over $2f + 1$ identical commit messages, consensus is achieved. Then, the primary node shares the confirmation result with all nodes (including candidate nodes) and updates the scores of all nodes based on the formula in the previous section. Nodes in the candidate node set and consensus node set will be reorganized every 100 requests. The lowest scoring m nodes

will be removed from the consensus set and added to the end of the candidate list. The highest-scoring m nodes in the candidate set will be added to the consensus node set and renumbered accordingly. At this point, a consensus process concludes.

C. Identity Authentication Scheme Based on TRUG-PBFT

This section elucidates the process in three principal steps: 1) vehicle registration; 2) TRUG reselection; and 3) vehicle identity authentication. Note that TRUG refers to the consensus group mentioned earlier. The overall system architecture is depicted in Fig. 3, while Fig. 4 depicts the runtime sequence diagram of the system.

Step-1 (Vehicle Registration): All vehicles must undergo registration and obtain an ID before joining the blockchain network. The registration information will be inscribed on the blockchain for user identity verification. This study employs a ZKP algorithm to validate identities, with vehicle privacy data stored on the main chain and the public parameters of the ZKP on the secondary chain. The detailed registration process will be expounded in the subsequent section.

Steps 2.1 and 2.2 (TRUG Identification Request): The vehicle initiates an access request, which is received by the RSU and subsequently broadcast to all $RSU[n]$ within the service range of the BS (set to n). The primary node then solicits service from the BS. Upon receiving the request, the BS prepares a unique identification ID_{TRUG} for the TRUG and dispatches a consensus message to all $RSU[n]$ nodes within its range.

Steps 2.3–2.5 (TRUG Reselection and Storage): In the IoV, RSUs may include false or unauthorized nodes, making it imperative to use a consensus algorithm to select reliable

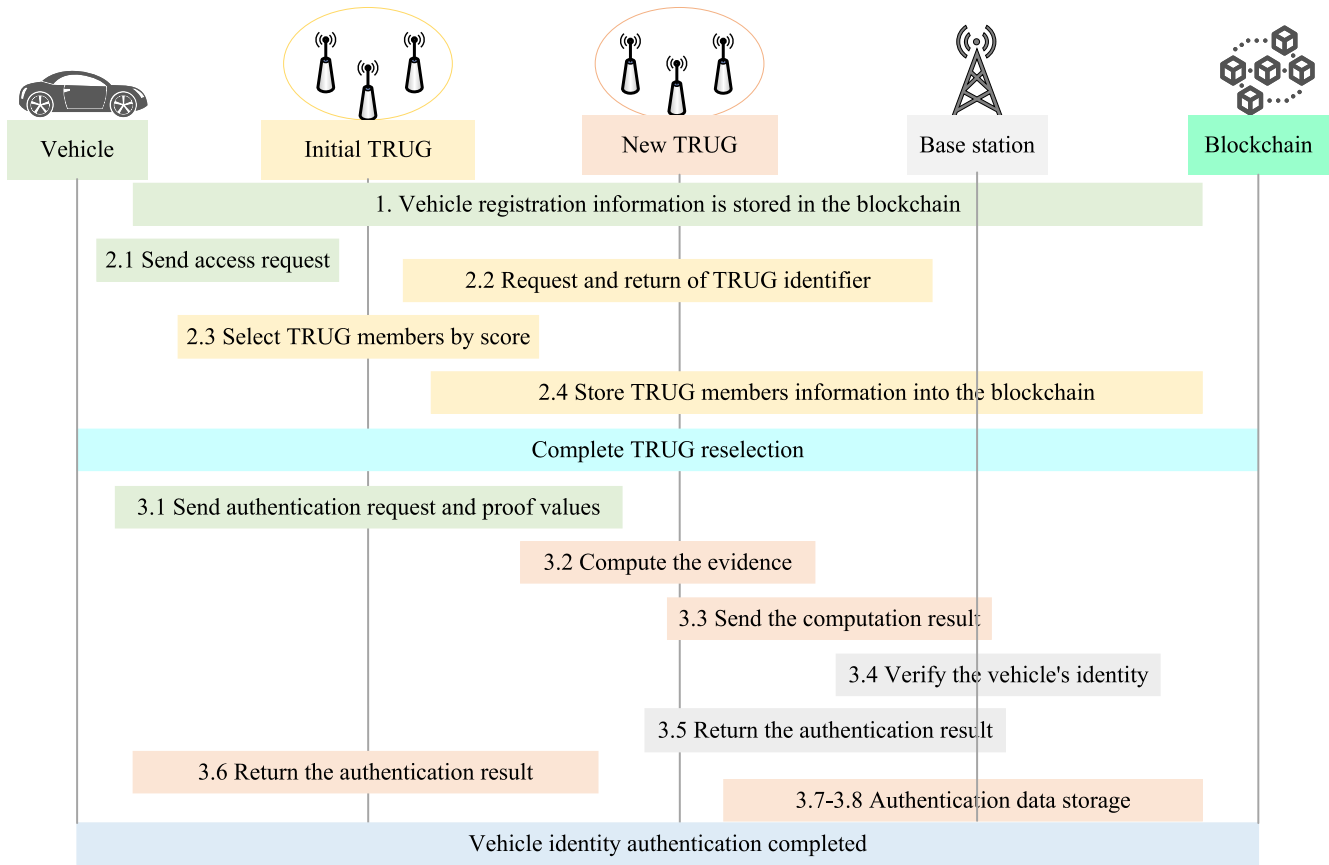


Fig. 4. System operation timing diagram.

and secure RSUs. The traditional PBFT consensus algorithm achieves network-wide consensus on the ledger but does not ensure the security and reliability of all network nodes. This study employs an optimized PBFT algorithm to evaluate RSU nodes, detailed in Algorithm 1. Through Algorithm 1, higher scoring RSU nodes are selected as TRUG members (i.e., the consensus group). The BS assigns the prepared ID_{TRUG} to the newly formed TRUG, and all member RSUs in the group are deemed trustworthy and share the identification ID_{TRUG} . The primary node then packages the TRUG member information into a block, and after TRUG-PBFT consensus, the block is added to the secondary chain. When a vehicle sends an access request to $RSU[i]$, $RSU[i]$ will return the TRUG member information to the vehicle.

Steps 3.1–3.6 (Vehicle Identity Authentication): Upon receiving the TRUG member information, the vehicle forwards identity authentication requests and ZKP values to the nearest TRUG member $RSU[j]$. $RSU[j]$ disseminates the proof values to other TRUG members, who calculate the values sent by the vehicle and transmit the results to the BS. The BS verifies the vehicle's identity and returns the authentication results to $RSU[j]$, which, in turn, relays the results to the vehicle. Depending on the authentication outcome, $RSU[j]$ considers whether to provide services to the vehicle. Detailed descriptions of the authentication algorithm will be provided in Section IV of this article.

Considering the short intervals between vehicle access requests, RSUs within the BS's range will conduct multiple rounds of consensus grouping in a short period, significantly reducing the efficiency of vehicle authentication. In reality, the probability of trusted RSU nodes becoming untrusted within a short time is small. Therefore, TRUG reselection is performed every Z times a vehicle requests access. Simply put, only when the number of vehicle requests is a multiple of Z will step 2 (TRUG identification request, TRUG reselection, and storage) be executed. For nonmultiple times of Z vehicle requests, vehicles only undergo step 3 (vehicle identity authentication). To prevent malicious attacks by unauthorized entities gaining knowledge of parameter Z , Z is set as a variable, dynamically adjusted based on the number of untrusted RSU nodes appearing within a unit of time, with an initial value of 100.

Steps 3.7 and 3.8 (Vehicle Authentication Data Storage): The primary node in the TRUG packages authentication data (such as authentication results) into blocks, which are added to the secondary chain through TRUG-PBFT consensus by the group's nodes. Consequently, all $RSU[n]$ within the BS's range store the vehicle's authentication results. When a vehicle moves into the coverage area of a specific $RSU[n]$, it presents its authentication, and the $RSU[n]$ quickly verifies it against the stored authentication results. If the verification is successful, $RSU[n]$ provides traffic information services to the vehicle, which seamlessly transitions to the next RSU. After

each consensus round, node scores are updated, laying the groundwork for TRUG member reselection.

IV. IDENTITY AUTHENTICATION BASED ON ZERO-KNOWLEDGE PROOF

This article uses a ZKP scheme based on lattice. The computational problem in the lattice is the inhomogeneous small integer solution (ISIS) problem, which is defined as follows.

Given a modulus q , a matrix $A \in \mathbb{Z}_q^{n \times m}$, its column $a_i \in \mathbb{Z}_q^n$, and a vector $y \in \mathbb{Z}_q^n$, find an integer vector $x \in \mathbb{Z}_q^m$ that satisfies

$$Ax = \sum_i a_i x_i = y \in \mathbb{Z}_q^n. \quad (8)$$

A. Initialization Phase

Before a vehicle hits the road, it must be registered at the transportation department, which assigns a vehicle identifier and communication key, denoted as $\langle \text{Tag}, sk \rangle$. The *Tag* encompasses the vehicle's brand, chassis number, and other privacy-sensitive information. The $sk \in \mathbb{Z}_q^m$ is primarily used for encrypting information during the authentication process and is considered part of the vehicle's privacy. This information must not be revealed during the verification process. In case of malicious access, the transportation department can trace the vehicle through its *Tag*.

Simultaneously, the transportation department's cloud server generates P_σ , A , and y for each vehicle. A and y satisfy the condition $A \cdot \text{Tag} = y \pmod{q}$. P_σ is an $m \times m$ matrix comprising only 0 and 1, where A and y are publicly available. The vehicle will store $\langle \text{Tag}, sk, A, P_\sigma \rangle$, while the main chain stores $\langle \text{Tag}, A, P_\sigma, y \rangle$, and the secondary chain stores $\langle A, P_\sigma, y \rangle$.

B. Verification Phase

Via TRUG-PBFT consensus algorithm, a consensus group is selected within RSUs. These RSUs within the consensus group act as verification servers for ZKP. Assuming the number of verification servers is denoted as I , $i \in \{1, 2, \dots, I\}$. The validation algorithm process is depicted in Fig. 5.

Step-1: When a vehicle initially enters the BS's range, it sends a validation request to the nearest verification server i . The verification server i selects a random number $\alpha \in \mathbb{Z}_q$ and forwards it to the vehicle.

Step-2: The vehicle generates proof values, namely, the zero-knowledge evidence. Initially, the vehicle creates a random number x , where x represents an m -bit binary number, following the formula:

$$x = \{0, 1\}^m. \quad (9)$$

Then, based on the random number α sent by verification server i , compute the following four proof values:

$$\begin{cases} \beta = P_\sigma(sk + \text{Tag} \cdot \alpha) \\ c_1 = P_\sigma(\text{Tag} + x) \\ c_2 = H_1(c_1, A \cdot sk) \\ c_3 = A \cdot sk + \text{Tag}. \end{cases} \quad (10)$$

The vehicle transmits $m_1 = \langle \beta, c_1, c_2, c_3 \rangle$ to verification server i , and the verification server i forwards $m'_1 = \langle \alpha, \beta \rangle$ to other verifiers.

Step-3: The verification server computes the ZKP. Other verification servers j calculate c_{j4} based on the received α and β

$$c_{j4} = AP_\sigma^{-1}\beta - y\alpha. \quad (11)$$

Verification server i transmits $m_2 = \langle \beta, c_1, c_2, c_{i4} \rangle$ to the BS, while the other verification servers send $m'_2 = \langle c_{j4} \rangle$ to the BS.

Step-4: The BS authenticates the vehicle's identity. The BS aggregates the data sent by multiple verification servers, establishing $C_4 = \{c_{14}, c_{24}, \dots, c_{j4}\}$. If the vehicle's identity is legitimate and the initialization is completed as required by the regulatory department, for each vector in C_4 , it holds that $c_2 = H_1(c_1, c_{j4})$, thus validating the vehicle's identity. If the results in C_4 align with expectations, the BS further authenticates the vehicle's identity using the following formula:

$$c_{j5} = c_3 - c_{j4}. \quad (12)$$

The BS aggregates and computes the data, establishing $C_5 = \{c_{15}, c_{25}, \dots, c_{j5}\}$. If each vector in C_5 is identical, and the corresponding *Tag* is confirmed to exist in the main chain, it further validates the legitimacy of the vehicle's identity. Otherwise, the vehicle's identity is deemed invalid.

Once the BS confirms the legitimacy of the vehicle's identity, it generates c_6 as evidence through (13) to authenticate the legitimacy of the BS's identity

$$c_6 = P_\sigma^{-1}(\beta - c_{14}\alpha). \quad (13)$$

Then, the BS sends $m_3 = \langle \text{CON}, H_2(c_6) \rangle$ to the verification server, where CON represents the conclusion drawn by the BS regarding the legitimacy of the vehicle's identity.

Step-5: The verification server applies the TRUG-PBFT consensus algorithm to the verification result. Upon successful consensus, it records the verification conclusion on the auxiliary chain. If CON indicates the legitimacy of the vehicle's identity, the verification server forwards $m_4 = \langle H_2(c_6) \rangle$ to the vehicle.

Step-6: The vehicle assesses the legitimacy of the BS's identity. It computes $H_2(sk - \alpha x)$ and compares it with $H_2(c_6)$. If they match, it confirms the legitimacy of the BS's identity.

During the initialization phase, the parameters $\langle \text{Tag}, sk, P_\sigma, A, y \rangle$ are generated by the cloud server of the traffic management department, eliminating the need for the vehicle or verification server's computational resources. Additionally, both the BS and verification servers precalculate the value of AP_σ^{-1} .

V. SECURITY ANALYSIS

A. Privacy of Location

Within this scheme, the generated values β, c_1, c_2, c_3 by the vehicles are outputs of a pseudo-random number generator and a one-way hash function. They are not generated based on the vehicle's location privacy. Moreover, these values are random, distinct, and independent. Adversaries cannot differentiate whether messages originate from the same vehicle, rendering the location indistinguishable.

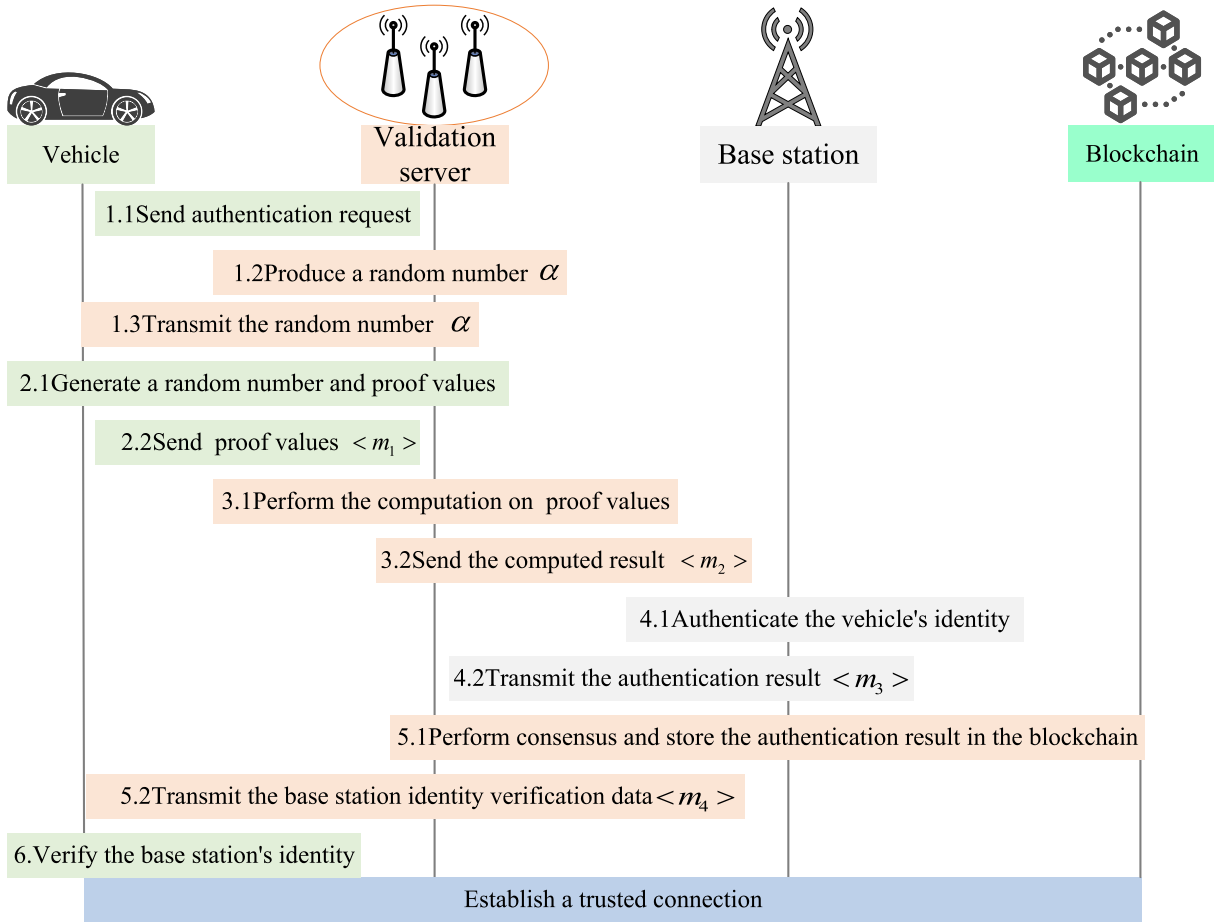


Fig. 5. Authentication process.

B. Unforgeability

Vehicles undergo identity verification by generating β, c_1, c_2, c_3 . These proof values are computed using the vehicle's Tag, sk, P_σ, A, x , etc. Since hash functions, random value x , and key sk are utilized in the scheme, attackers cannot access authentic values. Even if leaked, these values cannot be used in another session because each session computes values differently. Additionally, no existing machine or attacker can solve the ISIS problem, forming the basis of this authentication scheme, with a neglectable probability. Consequently, any attacker cannot extract the vehicle's Tag from leaked values.

C. Forward Secrecy

Forward secrecy entails that no attacker can extract previous information from all disclosed transactions. In each session, we randomly select session keys like α and x . The secret key sk is known only to the vehicle itself, while A and y are shared keys among the vehicle, verification servers, and the BS. Even if an attacker gains access to all transmitted messages, they cannot acquire the secret key. The only chance of accessing secret or session keys involves solving the ISIS problem, which even on quantum computers, cannot be solved in polynomial time. Therefore, attackers cannot access secret or session keys, ensuring forward secrecy in this scheme.

D. Resistance Against Man-in-the-Middle Attacks

Communication between vehicles and RSUs is insecure, whereas communication between RSUs and the BS is secure. Assuming an attacker eavesdrops on the communication, they might access some values within the transaction. However, if the attacker intends to alter and transmit these values to the vehicle/BS, they will not succeed. Proof messages are computed using $x \in \{0, 1\}^m$ randomly generated for each session and the secret key sk . Furthermore, each vehicle has a unique identification number (Tag). In this context, if an attacker generates any values to substitute the used ones, the authentication process would fail. The security of the proposed scheme relies on the ISIS problem, featuring unique identifier and secret key. Therefore, the only chance for attackers to execute such attacks is by solving the ISIS problem. We know that even in quantum computers, there is no way to solve this problem in polynomial time. Hence, the proposed scheme possesses resistance against MITM attacks, as attackers cannot disrupt transmitted messages through eavesdropping or alteration.

E. Resistance Against Replay Attacks

Suppose an eavesdropper transmits proof values β, c_1, c_2, c_3 obtained from a previous execution to the verification server, masquerading as the vehicle. These values are transferred

to the BS by adding a random value α , generated by the verification server. Since the verification server randomly generates α , the values of c_1 and c_5 will not match at the BS. Consequently, the eavesdropper cannot replay this scheme.

F. Resistance Against Impersonation Attacks

An attacker needs to compute β, c_1, c_2, c_3 to impersonate a vehicle. However, vehicles calculate these proof values using Tag, sk, P_σ, A, x . Similarly, when an attacker eavesdrops and obtains proof values to impersonate the BS, the attacker must impersonate the BS's response to the vehicle's proof. Yet, the BS possesses knowledge of the vehicle's P_σ, A, y and identification number Tag . For the attacker to succeed in impersonating, they must solve the ISIS problem. Therefore, the proposed scheme demonstrates resistance against impersonation attacks.

VI. PERFORMANCE ANALYSIS

The proposed approach primarily relies on an improved consensus algorithm and a ZKP-based authentication mechanism. Thus, to illustrate the performance advantages of the scheme, this section will initially conduct separate calculations and simulated analyses to evaluate the efficiency enhancements of the improved TRUG-PBFT consensus algorithm. Subsequently, it will compute the time expenses for authentication and simulate the authentication process, demonstrating the feasibility of the authentication approach proposed in this article from the perspective of authentication response time.

A. Simulation and Analysis of TRUG-PBFT Consensus Algorithm

Based on NS3 and utilizing C++, simulation of the TRUG-PBFT algorithm was performed, contrasting and analyzing it against PBFT and SG-PBFT [18] algorithms. This analysis tested the communication volume, throughput, and consensus latency of TRUG-PBFT across varying node quantities.

Throughput represents the number of transactions processed by a system within a unit of time, often measured in transaction per second (TPS) in blockchain systems, serving as a crucial performance metric

$$TPS = \text{Transaction} / \Delta t. \quad (14)$$

Consensus latency denotes the time taken for a transaction to achieve consensus across the entire network after submission, serving as another vital performance indicator.

During the experimental process, the number of consensus nodes in the TRUG-PBFT algorithm always remains at $N/3$ of the total node count. We have chosen to round down, resulting in consensus node counts of 6, 10, 13, 16, 20, 23, 26, 30, and 33. As depicted in Figs. 6–8, with an increase in RSU quantity, throughput gradually starts to decrease, while consensus latency progressively increases, accompanied by a continual rise in communication instances. This trend arises due to the escalating communication overhead with an increased RSU count, prolonging the time required for RSUs to reach consensus and consequently reducing the volume of transactions processed within a unit of time. Relative to

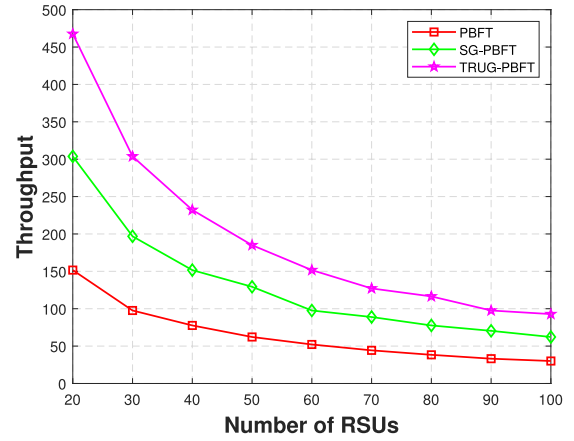


Fig. 6. Comparison of throughput.

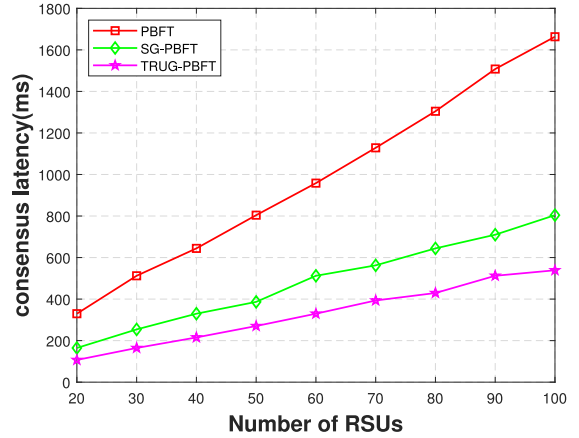


Fig. 7. Comparison of consensus latency.

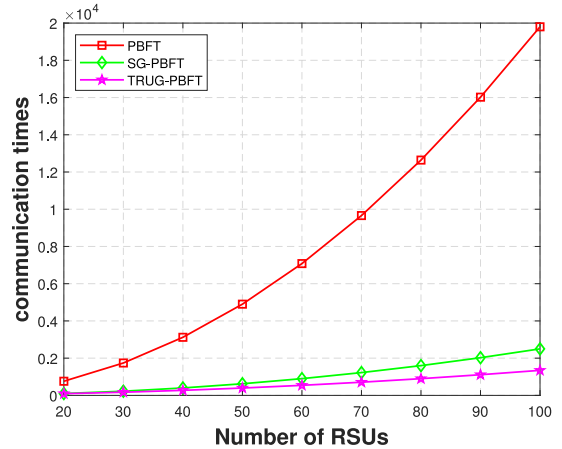


Fig. 8. Comparison of communication times.

traditional PBFT and SG-PBFT consensus mechanisms, the proposed TRUG-PBFT consensus, by limiting the number of consensus nodes, achieves higher transaction throughput and lower latency, enhancing consensus efficiency.

B. Assessment of Authentication Performance

1) *Authentication Overhead*: This section will calculate the performance overhead of the authentication scheme and

TABLE III
VARIOUS TYPES OF ARITHMETIC TIME OVERHEAD

Arithmetic type	The time consumption(ms)
Random number generation(T_r)	0.0185
Hash operation(T_h)	0.0495
Matrix-vector multiplication(T_m)	1.406
Scalar-vector multiplication(T_s)	0.0151

TABLE IV
AUTHENTICATION TIME CONSUMPTION

Authentication algorithm execution times	Time consumption (ms)
100	708.6
200	1426.7
300	2114.3
400	2847.4

compare it with other schemes. According to [23], parameters are set as follows: m is set to 2048, n is set to 64, modulus q is 257, and the hash function output is 256 bits. Simulation was conducted on a system configured with an i5-6500 processor, 16 GB of memory, running a Windows operating system.

Under the aforementioned conditions, this study simulated vehicle authentication with selected consensus RSUs and the BS, recording the overhead generated by lattice-based ZKP during the authentication phase. When vehicles begin authentication, multiple arithmetic operations are performed. Considering that the computational expense of addition and subtraction calculations is insignificant, they can be disregarded. The time costs of computing each arithmetic operation in a Python environment are detailed in Table III. The total computational overhead in the authentication scheme is calculated as $T_r + 3T_h + 5T_m + 4T_s \approx 7.2574(\text{ms})$. To precisely measure the algorithm's consumption, the authentication algorithm was executed 100, 200, 300, and 400 times separately, and the average time consumption was calculated, as shown in Table IV. From these tables, it can be concluded that the primary time-consuming operation during vehicle authentication is matrix-vector multiplication.

2) *Comparison of Authentication Schemes*: In order to analyze the superiority of our proposed solution, this article selected multiple authentication schemes for a comparative security evaluation. Additionally, it contrasts the authentication expenses of two identity verification schemes based on ZKP [6], [7].

Regarding the security assessment, Table V exhibits the fulfillment of specific properties, where $\checkmark$ denotes compliance with the respective characteristic, while $\times$ indicates a lack thereof. From the table, it is evident that our solution exhibits stronger security attributes. Schemes from [5], [6], and [7] display vulnerability in resisting MITM attacks. Both [5] and [17] fail to provide forward secrecy, allowing potential access by adversaries to past session's confidential information, thereby risking vehicle privacy disclosure.

In terms of performance evaluation, the primary time-consuming operation during our authentication process is the matrix-vector multiplication. In [6] and [7], the primary time-consuming operation involves modular exponentiation. During

the initialization phase, the AP_σ^{-1} operation is not immediately conducted but rather occurs during the offline process at the BS. Therefore, the time spent on this operation is not accounted for in the initialization phase.

As shown in Fig. 9, we conducted tests on various schemes in the registration phase for vehicle counts ranging from 10 to 90. During registration, the time cost increases linearly with the number of vehicles. We compared our scheme with those in [6] and [7]. As the number of vehicles increases, the time cost for all three schemes rises. The approach in [6] supports precomputing and preloading data for each entity in the registration phase, which is an isolated process, leading to reduced communication overhead and latency compared to the protocol in [7]. Our scheme requires less data in the registration phase, and the most time-consuming operation occurs offline at the BS. Consequently, the time expenditure during the registration phase is minimal. As illustrated in Fig. 10, our scheme not only ensures user privacy but also exhibits the best performance among the three authentication schemes. While the other two schemes involve numerous modular exponentiation operations during authentication, our scheme, although involving time-consuming matrix-vector multiplication, performs fewer operations compared to the other two, providing a certain superiority in authentication costs.

3) *Simulation Analysis*: In the previous section, factors, such as blockchain consensus latency and information dissemination time, were not considered. Only the time required for arithmetic operations in the authentication process was compared with other schemes. However, in this section, we will use simulation tools to analyze the network performance of the proposed scheme. To measure the network performance of our proposed protocol, we used Veins [24]. Veins is an open-source framework for simulating vehicular networks. It is based on two mature simulators: OMNeT++ (an event-based network simulator) and SUMO (a road traffic simulator). Table VI presents the parameter settings for this simulation.

In this article, we consider the interval between vehicle message transmission and RSU message processing as the transmission delay. We set up the simulation scenario described above, where vehicles periodically send messages according to our settings, and RSUs receive and process these messages. From Fig. 11, it can be observed that the number of vehicles affects the transmission delay, while variations in vehicle speed have little impact on the transmission delay. Subsequently, through simulation, we tested the packet loss rate, as shown in Fig. 12. As the number of vehicles gradually increases, the packet loss rate also increases, and vehicle speed also has a certain influence on the packet loss rate. Furthermore, we observed a sudden jump in the packet loss rate when the vehicle count attained 100. We surmise that when the number of vehicles reaches 100, it reaches the critical threshold of the RSU network bandwidth, leading to a sudden spike in the packet loss rate. Subsequently, after the number of vehicles surpasses 100, the packet loss rate exhibits a gradual and steady increase. This phenomenon is attributed to the adaptive transmission strategy, which is based on the IEEE 802.11p protocol. This strategy dynamically

TABLE V
AUTHENTICATION SECURITY COMPARISON

Scheme	Location privacy	Unforgeability	Forward secrecy	Resist man-in-the-middle attacks	Resist replay attacks	Resist impersonation attacks
[6]	✓	✓	✓	×	✓	✓
[7]	✓	✓	✓	×	✓	✓
[5]	✓	✓	×	×	✓	✓
[17]	✓	✓	×	✓	✓	✓
Our approach	✓	✓	✓	✓	✓	✓

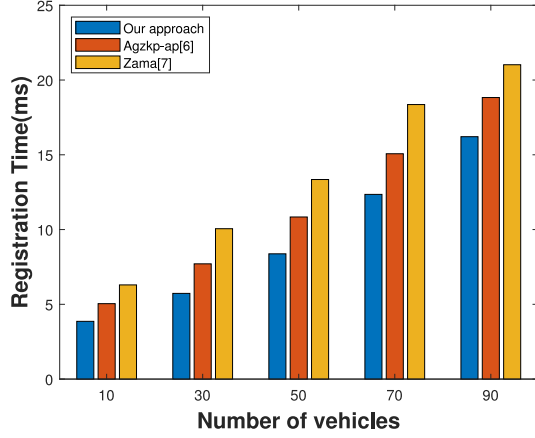


Fig. 9. Average registration time cost.

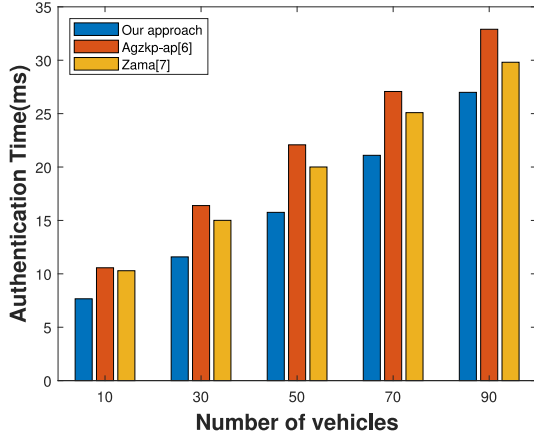


Fig. 10. Average authentication time cost.

adjusts the transmission rate based on channel conditions and the current network status, thereby mitigating collisions and retransmissions, and consequently reducing the packet loss rate.

Finally, this article compares the PBFT consensus with the TRUG-PBFT consensus scheme. Assuming there are m RSUs within the coverage area of the BS, and there are n vehicles within the BS's coverage area. In the first set of simulations, parameters are set as follows: 1) $m = 20$ and 2) $n \in [20, 80]$, and the results are shown in Fig. 13. In the second set of simulations, parameters are set as follows: 1) $m \in [20, 80]$ and 2) $n = 20$, and the results are shown in Fig. 14. According to Fig. 13, we can observe that with the increase in the number of vehicles, the average response time

TABLE VI
SIMULATION PARAMETERS CONFIGURATION

Parameters	Values
Simulation area	$2700 \times 2500(m^2)$
Data Transmission Rate	6Mbps
Noise level	-98dBm
Used channel	CCH
Transmission Power	20mW
Number of vehicles	10-200
Distance between vehicles	3m
Acceleration speed of vehicle	$0.8 \frac{m}{s^2}$
Deceleration speed of vehicle	$4.5 \frac{m}{s^2}$
Beacon interval	1s
Vehicle speed	30, 60, 90, 120 $\frac{km}{h}$
Simulation time	200s

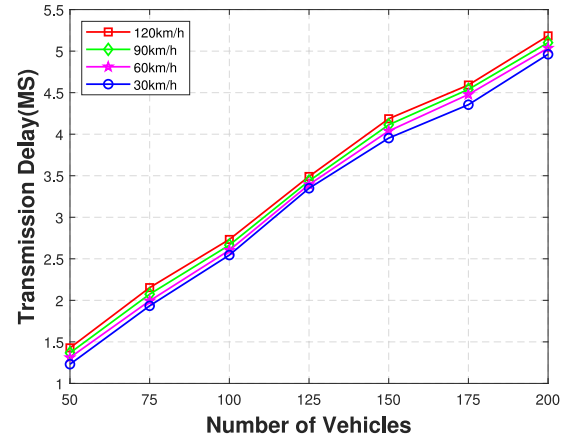


Fig. 11. Impact of speed on transmission delay.

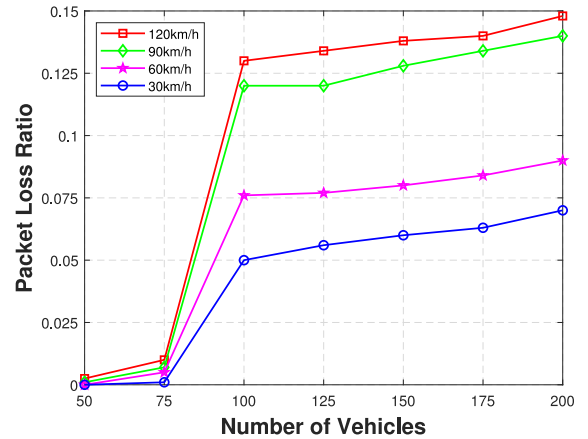


Fig. 12. Impact of speed on packet loss rate.

for authentication shows an upward trend. This is because the increase in the number of vehicles leads to an extension of the waiting time. Moreover, compared to the approach proposed

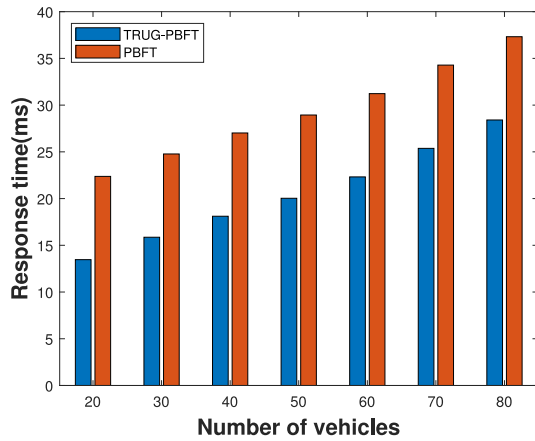


Fig. 13. Variation in authentication response time with number of vehicles.

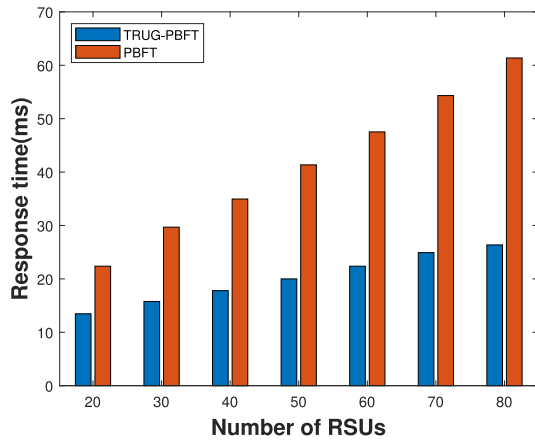


Fig. 14. Variation in authentication response time with number of RSUs.

in this article, the authentication response time of the PBFT scheme is longer. This is because the PBFT scheme requires longer consensus time, which to some extent leads to an increase in response time. Furthermore, by further observing Fig. 14, we find that with the increase in the number of RSUs, the average authentication response time also increases. This is because the increase in the number of RSUs leads to an increase in the time required for the BS to verify c_5 , and the time for RSUs to reach consensus also increases accordingly. Ultimately, these factors together contribute to an upward trend in the average authentication response time.

Simulation results show that the proposed scheme achieves authentication response times in the millisecond range for both single-user and multiuser access scenarios, making it feasible for practical applications. Moreover, compared to the PBFT consensus scenario, the proposed scheme has lower time consumption, making it suitable for vehicular networks.

VII. CONCLUSION

With the rapid development of the IoV, various security issues have emerged alongside the convenience it brings to people's travels. Identity authentication, as the first line of defense in IoV security, is crucial. Balancing authentication efficiency with protecting vehicle privacy is a research hotspot.

This article addresses the efficiency of authentication by proposing a score-based consensus grouping scheme. By optimizing the PBFT algorithm and implementing a score filtering mechanism, trusted consensus groups are dynamically generated in the IoV environment. The authentication results of vehicles are transmitted and shared among TRUG members, allowing all members to recognize the user. This reduces the number of authentications of vehicles by TRUG members, significantly improving authentication efficiency and enhancing user experience. Furthermore, to address vehicle privacy concerns, this article adopts a ZKP approach to protect vehicle privacy. Through security analysis and simulation experiments, the proposed method significantly improves verification efficiency without compromising vehicle privacy.

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